EXERCISE 5

I. Choose the best answer.

1. _____ is widely used for prediction

   a) regression analysis b) correlation analysis

   c) analysis of variance d) analysis of covariance

2. The linear regression analysis can be classified in to

   a) 4 types b) 3 types c) 2 types d) none of the above

3. The linear equation $Y = a + bx$ is called as regression equation of

   a) X on Y b) Y on X c) between X and Y d) 'a' on 'b'

4. In regression equation $X = a + by + e$ is

   a) correlation coefficient of Y on X b) correlation coefficient of X on Y

   c) regression coefficient of Y on X d) regression coefficient of X on Y

5. b_{YX} =

   a) $r_{XY}\frac{SD(X)}{SD(Y)}$ b) $r_{XY}\frac{SD(Y)}{SD(X)}$ c) $\frac{SD(X)}{SD(Y)}$ d) $\frac{SD(Y)}{SD(X)}$

6. If $b_{XY} > 1$ then b_{YX} is

   a) 1 b) 0 c) > 1 d) < 1

7. In the Regression equation $\hat{Y} - \bar{y} = r_{XY}\frac{SD(Y)}{SD(X)}\left(x - \bar{x}\right)$, $r_{XY}\frac{SD(Y)}{SD(X)}$ is

   a) b_{YX} b) b_{XY} c) r_{XY} d) $\text{cov}(X, Y)$

8. Using the regression coefficients we can calculate

   a) $\text{cov}(X, Y)$ b) $SD(X)$

   c) correlation coefficient d) coefficient of variance

9. Arithmetic mean of the regression coefficients b_{XY} and b_{YX} is

   a) $> r_{XY}$ b) $\geq r_{XY}$ c) $\leq r_{XY}$ d) $< r_{XY}$

10. Regression analysis helps in establishing a functional relationship between _____ variables.

    a) 2 or more variables b) 2 variables

    c) 3 variables d) none of these

11. _____ is the Father of mental tests

    a) R.A. Fisher b) Croxton and Cowden

    c) Francis Galton d) A.L. Bowley

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